1.2 Atoms, Molecules & Stoichiometry

Question Paper

Course	CIE A Level Chemistry
Section	1. Physical Chemistry
Topic	1.2 Atoms, Molecules & Stoichiometry
Difficulty	Hard

Time allowed: 20

Score: /10

Percentage: /100

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Question 1

1.00 g of carbon is combusted in a limited supply of oxygen. 0.50 g of the carbon combusts to form CO_2 and 0.50 g of the carbon combusts to form CO.

These were passed through excess NaOH (aq) and the remaining gas then dried and collected.

Assuming that all gas volumes were taken at 25˚C and 1 atm pressure, what was the volume of gas at the end of the reaction?

A. 3 dm^3

B. 2 dm^3

C. 1.5 dm^3

D. 1 dm^3

[1 mark]

Question 2

Egg shells make up 5% of the mass of chicken eggs. The average egg has a mass of 50 g.

Assume that chicken eggshell is pure calcium carbonate.

How many complete chicken eggshells would be needed to neutralise 50 cm^3 of 2.0 mol dm^{-3} ethanoic acid?

A. 4

B. 3

C. 2

D. 1

[1 mark]

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Question 3

Use of the data booklet is relevant to this question

When a sample potassium oxide, K_2O, is dissolved in 250 cm^3 of distilled water. 25 cm^3 of this solution is titrated against sulfuric acid with concentration of 2.00 mol dm^{-3}. Complete neutralisation takes place with 15 cm^3 of sulfuric acid.

What is the mass of the original sample of potassium oxide dissolved in 250 cm^3 of distilled water?

A. 56.6 g

B. 47.1 g

C. 28.3 g

D. 2.83 g

[1 mark]

Question 4

Use of the data booklet is relevant to this question

Iron and chromium can be made into an alloy called ferrochrome. Ferrochrome can be dissolved in dilute sulfuric acid to produce $FeSO_4$ and $Cr_2(SO_4)_3$. The $FeSO_4$ reacts with $K_2Cr_2O_7$ in acid shown in this equation:

$$14H^+ + 6Fe^{2+} + Cr_2O_7^{2-} \rightarrow 2Cr^{3+} + 6Fe^{3+} + 7H_2O$$

1.00g of ferrochrome is dissolved in dilute sulfuric acid, and then titrated, 13.1cm^3 of 0.100 mol dm^{-3} $K_2Cr_2O_7$ is needed for the complete reaction.

In the sample of ferrochrome, what is the percentage by mass of *Fe*?

A. 43.9

B. 12.2

C. 4.39

D. 1.22

[1 mark]

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Question 5

10 cm^3 of methane and 10 cm^3 of ethane were sparked with an excess of oxygen. Once cooled the remaining gas was passed through aqueous potassium hydroxide.

Assuming all measurements were taken at 25˚C and 1 atm pressure.

What volume of gas is absorbed by the alkali?

A. 45 cm^3

B. 30 cm^3

C. 20 cm^3

D. 10 cm^3

[1 mark]

Question 6

A solution of Sn^{2+} ions will reduce MnO_4^- ions to Mn^{2+} ions when acidified. The Sn^{2+} ions are oxidised to Sn^{4+} ions in this reaction.

How many moles of Mn^{2+} ions are formed when a solution containing 9.5 g of $SnCl_2$
(M_r: 190) is added to an excess of acidified $KMnO_4$ solution?

A. 0.010

B. 0.015

C. 0.020

D. 0.050

[1 mark]

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Question 7

Use of the data booklet is relevant to this question

Some fireworks can use the reaction between aluminium powder and anhydrous barium nitrate as a propellant. Metal oxides and nitrogen are the only products.

$$10Al + 3Ba(NO_3)_2 \rightarrow 5Al_2O_3 + 3BaO + 3N_2$$

When 0.783 g of anhydrous barium nitrate reacts with an excess of aluminium what is the volume of nitrogen produced?

A. 144.0 cm^3

B. 93.6 cm^3

C. 1.62 cm^3

D. 72.0 cm^3

[1 mark]

Question 8

Use of the data booklet is relevant to this question

Excess acidified aqueous potassium dichromate (VI) was mixed with 2.76 g of ethanol. The reaction mixture was then boiled under reflux for one hour. The organic product was collected by distillation.

The yield of the product was 75.0%

What is the mass of the product collected?

A. 1.98 g

B. 2.70 g

C. 2.07 g

D. 4.80 g

[1 mark]

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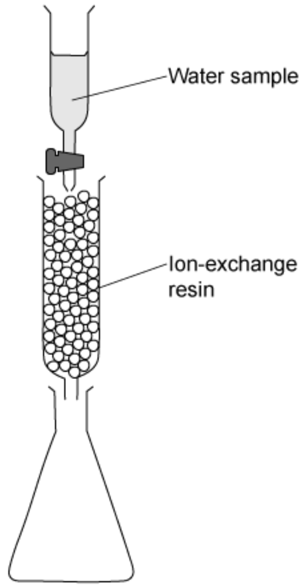
Question 9

The concentration of calcium ions in a sample of water can be determined by using an ion-exchange column, shown in the diagram below:



A 50 cm^3 sample of water was passed through the ion-exchange resin, this contained dissolved calcium sulfate

Each calcium ion in the sample was exchanged for two hydrogen ions. The resulting acidic solution collected in the flask required 25 cm^3 of 1.0×10^{-2} mol dm^{-3} potassium hydroxide for complete neutralisation.

What was the concentration of the calcium sulfate in the original sample?

A. 4.0×10^{-2} mol dm^{-3}

B. 2.0×10^{-2} mol dm^{-3}

C. 1.0×10^{-2} mol dm^{-3}

D. 2.5×10^{-3} mol dm^{-3}

[1 mark]

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Question 10

Bromide ions are oxidised to bromine by lead (IV) chloride. The Pb^{4+} ions are reduced to Pb^{2+} ions in this reaction.

If 6.980 g of lead (IV) chloride is added to an excess of sodium bromide solution, what mass of bromine would be produced?

A. 3.196 g

B. 6.392 g

C. 0.799 g

D. 1598 g

[1 mark]

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Model Answers: Hard

1

The correct answer is **D** because:

	Steps	Calculation
1	Reactions taking place during combustion	$C + O_2 \rightarrow CO_2$ $2C + O_2 \rightarrow 2CO$
2	Reactions taking place when passed through NaOH	$CO_2 + 2NaOH \rightarrow Na_2CO_3 + H_2O$ CO + NaOH – No reaction
3	**Moles** of carbon monoxide *Moles = mass / M_r*	0.5 / 12 = 0.0417
4	**Volume** of gas remaining Rearrange the equation: *Amount of gas (mol) =* $\frac{\text{Volume of gas (dm}^3\text{)}}{24}$	0.0417 = x / 24 0.0417 x 24 = **1.0008 dm^3**

2

The correct answer is **C** because:

	Steps	Calculation
1	Reactions taking place during neutralisation	$2CH_3COOH + CaCO_3 \rightarrow (CH_3COO)_2Ca + CO_2 + H_2O$
2	Find the **mass** of shell of an average egg	5% of 50 g

		$0.05 \times 50 = 2.5$ g
3	Find the **moles** of calcium carbonate *Moles = mass / M_r*	$2.5 / 100 = 0.025$
4	Find the **moles** of ethanoic acid	50 cm^3 of 2 mol dm^{-3} = $(50 \times 2) / 1000$ = 0.1 mol
5	2 moles of ethanoic acid reacts with 1 mole of calcium carbonate 0.05 mole of calcium carbonate would be needed to react with 0.1 mol of ethanoic acid	$0.05 / 0.025$ = **2 egg shells needed**

3

The correct answer is **C** because:

Steps		Calculation
1	Reactions taking place	$K_2O + H_2O \rightarrow 2KOH$ $2KOH + H_2SO_4 \rightarrow K_2SO_4 + 2H_2O$
2	**Moles** of sulfuric acid used	Concentration x volume in dm^3 $2.00 \times 0.015 = 0.03$ mol
3	**Mole ratio** is 2:1	So 0.06 mol of KOH will react with 0.03 mol H_2SO_4
4	Moles in the **original solution** Moles in 25cm^3 of KOH = 0.06 This was a sample from the original solution of 250 cm^3	Moles in 250 cm^3 = $0.06 \times (250 / 25)$ = 0.6 mol

5	**Moles** of K_2O 1 mole of K_2O makes 2 moles of KOH	0.6 / 2 = 0.3 mol
6	**Mass** of K_2O *Mass = mol x* M_r	Mass = 0.3 x 94.2 = **28.26 g**

4

The correct answer is **A** because:

Steps		**Calculation**
1	Number of **moles** of $K_2Cr_2O_7$	13.1 x 0.1 / 1000 = 1.31×10^{-3} mol
2	**Mole ratio** is 1 mole of $K_2Cr_2O_7$ reacts with 6 moles of Fe	Moles of Fe = 1.31×10^{-3} x 6 = 7.86×10^{-3}
3	**Mass** of Fe in 1.00 g *Mass = mol x* M_r	Mass = 7.86×10^{-3} x 55.8 = 0.4386 g
4	**% by mass** of Fe in ferrochrome	0.4386 / 1.00 x 100 = 43.9%

5

The correct answer is **B** because:

Steps		**Calculation**
1	Combustion reactions	$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$ $C_2H_6 + 3.5O_2 \rightarrow 2CO_2 + 3H_2O$
		Moles of carbon dioxide produced

2	Once cooled the only product still as a **gas** is CO_2, this will react with the potassium hydroxide.	From the methane reaction 1 mole of CO_2

The correct answer is **B** because:

Steps		Calculation
1	Combustion reactions	$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$ $C_2H_6 + 3.5O_2 \rightarrow 2CO_2 + 3H_2O$
2	Once cooled the only product still as a **gas** is CO_2, this will react with the potassium hydroxide.	Moles of carbon dioxide produced From the methane reaction 1 mole of CO_2 From the ethane reaction 2 moles of CO_2
3	**Volume** of gas produced	Methane = 10 cm^3 Ethane = 20 cm^3 Total gas produced **30 cm^3**

6

The correct answer is **C** because:

Steps		Calculation
1	Reactions (from the data booklet)	$Sn^{2+} \rightarrow Sn^{4+} + 2e^-$ $MnO_4 + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$
2	Balance electrons in each equation	$5Sn^{2+} \rightarrow 5Sn^{4+} + 10e^-$ $2MnO_4 + 16H^+ + 10e^- \rightarrow 2Mn^{2+} + 8H_2O$
3	Add the two equations	$5Sn^{2+} + 2MnO_4 + 16H^+ \rightarrow 5Sn^{4+} + 2Mn^{2+} + 8H_2O$
	2 moles of MnO_4^- reacts with 5 moles of Sn^{2+}	Mol = mass / M_r x 0.4

4	or 0.4 MnO_4^- moles react with 1 mole of Sn^{2+}	= 9.5 / 190 x 0.4 = **0.02 mol**

7

The correct answer is **D** because:

Steps		Calculation
1	Moles of barium nitrate	M_r = 261.3 Moles = 0.783 / 261.3 = 0.0029
2	3 moles of barium nitrate produces 3 moles of Nitrogen *Volume cm^3 = moles x 24000*	Volume N_2 = 0.0029 x 24000 = **71.9 cm^3**

8

The correct answer is **B** because:

Steps		Calculation
1	Reaction	$CH_3CH_2OH \rightarrow CH_3COOH$
2	**Moles** of ethanol *Moles = mass / M_r*	M_r = 46 Moles = 2.76 / 46 = 0.06
3	**1: 1 ratio** so moles of CH_3COOH = 0.06	Mass of CH_3COOH (M_r 60) Mass = 0.06 x 60 = 3.6
4	75 % **yield**	75 / 100 x 3.6 = **2.7 g**

9

The correct answer is **D** because:

Steps		Calculation
1	Write out both reactions	$CaSO_4 + 2H^+ \rightarrow H_2SO_4$ $H_2SO_4 + 2KOH \rightarrow K_2SO_4 + 2H_2O$
2	**Moles** of potassium hydroxide *Moles = volume x concentration*	Volume = 0.025 dm^3 Moles = 0.025 x 1.0 x 10^{-2} = 0.00025 mol
3	**Mole ratio** 2 moles of potassium hydroxide with 1 sulfuric acid	Moles of sulfuric acid = 0.000125 mol
4	**Concentration** of Calcium sulfate *Concentration = moles / volume*	Concentration = 0.000125 / (50 / 1000) = **0.0025 or 2.5 x10^{-3} mol dm^{-3}**

10

The correct answer is **A** because:

Steps		Calculation
1	Reaction	$PbCl_4 + 2NaBr \rightarrow PbCl_2 + Br_2 + 2NaCl$
2	**Moles** of lead chloride *Moles = mass / M_r*	M_r = 349 Moles = 6.980 / 349 = 0.02 mol
	Ratio 1:1 so moles Br_2 = 0.02	

3	Mass of Br_2 *Mass = moles x M_r*	Mass = 0.02 x 159.8 = **3.196 g**